

D: $I_A = V/2R$

$V = 2(R + \frac{1}{2}) = 2(\frac{3R}{2} + \frac{1}{2}) = 3R + 1$

22 A $V = E - Ir$

When current $I = 0$, $V = E = 2.5V$

$$0.9 = 2.5 - 0.80 r$$

$$r = 2.0 \Omega \text{ or use gradient of graph} = -r$$

23 A Apply RH Grip Rule to determine the direction of B-field at O due to the currents flowing